TESTED AND HOPED ON

control software for the physical world, and the language that proves it

Nobody can prove a controller will never reach the network while it is steering a seven-ton flywheel. It is tested, reviewed, and hoped on.

Agents now write that software. The grid is where it bites first.

WHY NOW

Coding agents write production control code this year.

WHY THE GRID

Every new inverter is a software product under UL 1741 SB, and Utah's data centers are the load.

WHY HERE

Torus is scaling a factory in Salt Lake; Hill AFB verifies flight software by review.

MENTL

a language whose compiler proves what a program can never do

NEVER

"Never reaches the network, never allocates, never exceeds this bound": proven through every call, or the program does not build. The audit names it: Real-time safe, Sandbox.

WHY

Every fact carries its reason, walkable to the line that decided it. A dispatch explains itself.

LEARNS

A learning rule is read off the chain it learns on, inside a loop proven not to allocate.

THE SAME PROGRAM, ONE PRICE READ ADDED INSIDE THE REFLEX

REFUSED

```
fn decide(valid, hz, rocof, level, hour, rush, clean, out) -> Mode with !Alloc + !Network + !Filesystem = {
  ...
  let price = price_m("https://api.eia.gov/v2/electricity/rto/region-data/…")
  let pulse_ask = if rush == 1 { if price > 50000 { 300000 } else { 200000 } } else { … }
  ...
}
```

`decide` → `price_m` → `fetch_param` → `Network` the path the network takes into a function that swore it never would

```
effects: E_EffectMismatch error: effect row mismatch: !Alloc + !Network + !Filesystem vs Mutate + Memory + Trap + Network — at 107:1-228:2: declared
as decide; Network enters at examples/grid/station-leak:176:5 through `price_m` → at examples/grid/station-leak:73:14 through `fetch_param` → declared
as fetch_param, an op of Network declared at dsp/signal:66:1 at examples/grid/station-leak:107:1-228:2
effects: E_EffectUnhandled error: effect Network reaches the executable root with no absorbing handler — calling one of its ops would read garbage
evidence at runtime; nothing executes unproven. No handler in scope handles Network — declare one and install it with ~> over the performing chain.
declared as main; Network enters at examples/grid/station-leak:428:3 through `step` → at examples/grid/station-leak:409:16 through `decide` → at
examples/grid/station-leak:176:5 through `price_m` → at examples/grid/station-leak:73:14 through `fetch_param` → declared as fetch_param, an op of
Network declared at dsp/signal:66:1 at examples/grid/station-leak:427:1-430:2
```

This is what review cannot do: the build refuses, and names the line. Price belongs to the hour's plan, read before the reflex; a lookup inside the millisecond's decision is the bug.

Self-hosting: the pinned compiler reproduces itself byte for byte. One WebAssembly file. Open source. [[years]] of commits.

BUILT THIS WEEK

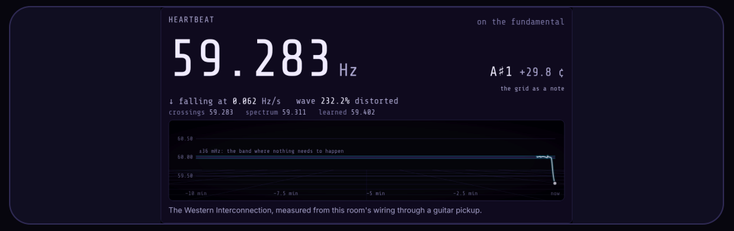
beside what existed before

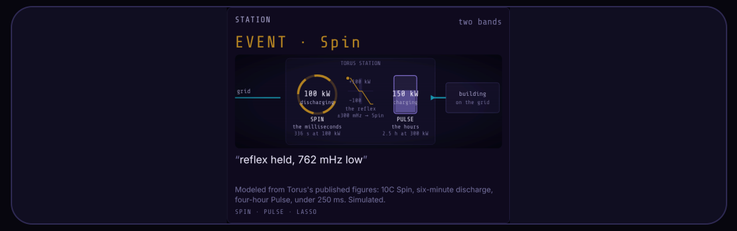
BEFORE KICKOFF

- Mentl: the compiler, the proofs, the course, the Wasatch piece made of Salt Lake's readings
- sonido: 36 effects, the plugin, the browser editor, the analysis crate, the pedal firmware

SINCE MONDAY 4 PM

- The ear: grid frequency from a guitar, three ways, to the millihertz, on the analysis crate
- The controller in Mentl, its proofs, its refusal twin, the learner
- The page, the director, the data, the real event replayed, the session recorded
- The gate's verdicts on proposed code: PROVEN, and REFUSED with the line; the pedal firmware as a second ear







./LEDGER.SH PRINTS EVERY COMMIT WITH ITS TIME

WHO NEEDS IT, IN UTAH

TORUS

Their Station hears every cycle. Their security is a perimeter. The program itself has never been provable, until now. The Torus judge heard this demo today.

HILL AFB, 309TH SOFTWARE ENGINEERING GROUP

Utah's largest employer of software developers, verifying flight software by review. A proof is a deliverable they can file.

EVERY UTAH COMPANY PUTTING AGENTS ON PRODUCTION CODE

A gate that answers PROVEN or REFUSED, with the line, before anything an agent wrote runs.

First hires: compiler and controls engineers, Salt Lake City.

WHAT $10,000 UNLOCKS

ninety days, two introductions, one co-founder

Day 30

The proven controller on inverter-class hardware on my bench. SAM.gov and SBIR.gov registered. Company formed. [[date]]

Day 60

One Utah site, one ear on a real Station-class inverter, the dispatch explained to its operator. The SBIR Phase I draft with Nucleus Grow.

Day 90

Ninety days of decisions on the page, the proposal filed, the first paid pilot signed.

The business: paid pilots with asset operators, a hosted proof gate for teams running coding agents, the SBIR lane for software assurance.

THE $10,000

$4,000 inverter-class hardware and a GPS clock
$3,000 ninety days of the proof gate hosted
$3,000 SBIR preparation and travel

What I need from you:
an introduction at Torus engineering
an introduction at the 309th
a co-founder with power-electronics depth

THE TRUSTED BASE

- The compiler is written in Mentl and the pinned binary reproduces itself byte for byte; that fixed point is the release.
- The verifier is sound for what it decides and incomplete by design: a claim it cannot decide is reported as pending, never silently passed.
- One false proof is known and documented: a recursion deep enough to exhaust the stack traps although its function says it cannot.
- Rust prevents memory bugs, not effects; Simulink and SCADE verify models and generate C. Neither states "never reaches the network" as an obligation through every callee, proves a function allocation-free, or derives a learning rule from the chain.

RUNNING ON A CONTROLLER

- The proof is at compile time. The artifact is one WebAssembly module importing WASI and nothing else.
- It runs wherever wasmtime runs, which includes the Linux supervisors that sit beside inverters today; the demo's controller runs that way and answers each reading in under a millisecond.
- A native backend is on the roadmap; the pedal in the demo runs the ear natively on a Cortex-M7 today.
- The learner keeps learning inside a loop proven not to allocate: learning at the edge that cannot stall.
- A running computation is a value: the compiler's own resume verb restores a persisted image and continues, so durability is construction, not infrastructure.

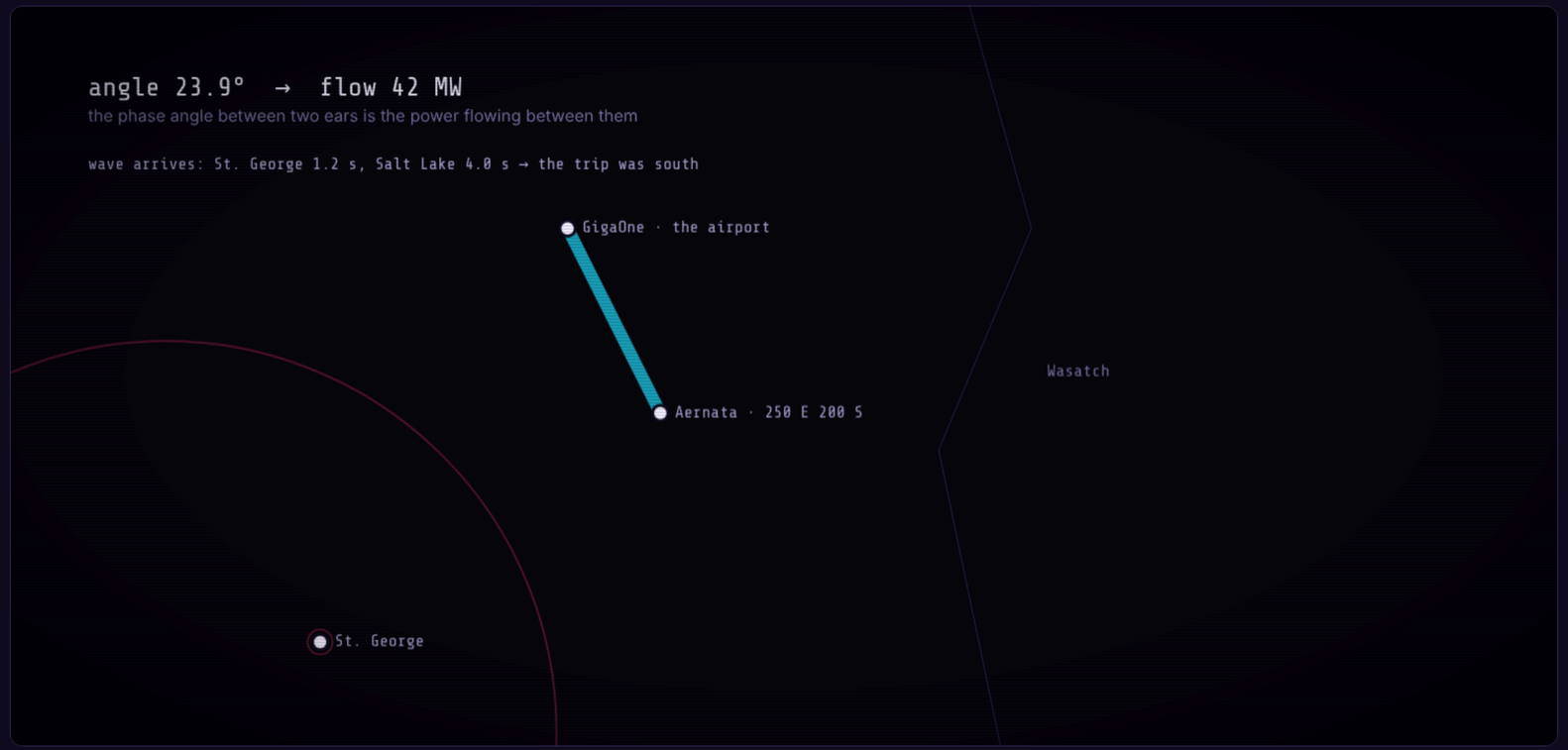
THE GRID THROUGH A DSP LIBRARY'S LENS

IN SONIDO	ON THE GRID
Pitch detection (the tuner)	Frequency: 60 Hz is a B, 49 cents flat; the 2019 event is a key change to A♯
THD (the distortion analyzer)	Power quality: the wave's timbre says what is plugged in
A compressor	Droop control: threshold, ratio, ceiling, hold; the controller's fast band is exactly that curve
A noise gate	Islanding detection
Tuning to a reference, the beat going to zero	Reconnecting an island: a synchroscope is a tuner
The correlation meter	Two sites: the angle between them is the flow
LMS (the adaptive filter)	The learner, written twice: by hand in Rust, and derived by the compiler in Mentl

THE MESH

MESH

SIMULATED



Two Stations with GPS time see the **angle** between them, which is the flow on the line between them, and an event's wave arrives at one before the other. FNET has done this from wall outlets since 2004; a fleet that both sees and answers at 10C has not existed. Simulated from the day's shape and the replayed event.